

GRE VPN

Generic Routing Encapsulation

Tunneling Protocol | GNS3 Lab Guide

PREPARED BY

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1. VPN FUNDAMENTALS

A Virtual Private Network (VPN) is a technology that enables the creation of secure, private communication channels over a shared or public network infrastructure. The primary objective of any VPN solution is to extend private network connectivity securely across a public medium such as the Internet, as if the communicating endpoints were directly connected on the same local network (so we can say that the main object of VPN is connect private networks over public network).

1.1 VPN Classification by Security Model

VPNs are broadly categorized into three security models, each differing in how they handle data confidentiality, authentication, and integrity:

1.1.1 Trusted VPN

Definition

A Trusted VPN provides privacy through logical separation of customer traffic by the service provider — without encrypting the data itself. Security is derived from trust in the provider's network infrastructure rather than cryptographic mechanisms.

Key Characteristics:

- Traffic isolation is achieved through logical segmentation (e.g., MPLS labels, VLANs).
- Encryption, authentication, and data integrity are NOT inherently enforced.
- Confidentiality depends on the provider's adherence to contractual SLAs.
- Suitable for enterprise deployments over trusted MPLS core networks.

1.1.2 Secured VPN

Definition

A Secured VPN protects data in transit using cryptographic mechanisms — encryption, authentication, and integrity verification — ensuring confidentiality and data protection even across fully untrusted networks such as the public Internet.

Key Characteristics:

- Encryption ensures data confidentiality (e.g., AES-256, 3DES).
- Authentication verifies the identity of communicating peers.
- Integrity mechanisms (e.g., SHA-256 HMAC) protect against tampering.
- Examples: IPsec VPN, SSL/TLS VPN, DMVPN, GRE over IPsec.

1.1.3 Hybrid VPN

Definition

A Hybrid VPN combines the logical traffic separation of a Trusted VPN with the cryptographic security mechanisms of a Secured VPN — providing both provider-based isolation and end-to-end encryption.

- Example: MPLS VPN carrying encrypted IPsec tunnels.
- Ideal for scenarios requiring both high-performance MPLS routing and data-level security.

1.2 VPN Classification by Connectivity Model

1.2.1 Site-to-Site VPN

A Site-to-Site VPN establishes a persistent, encrypted (or tunneled) connection between two or more geographically separated network sites over a public or private WAN. The tunnel is typically established between gateway routers or firewalls, and all hosts behind each site communicate transparently through it.

- Use Case: Head Office connecting to multiple Branch Offices.
- Example: CAIRO LAN (10.1.1.0/24) securely communicating with TANTA LAN (192.168.1.0/24) via GRE Tunnel.

1.2.2 Remote Access VPN

A Remote Access VPN allows individual users or endpoint devices to connect securely to a private corporate network from any remote location using the Internet. Authentication is performed per user, and a secure tunnel is established on demand.

- Use Case: Remote employee accessing internal company resources.
- Common Protocols: SSL VPN (Cisco AnyConnect), IPsec IKEv2, OpenVPN, WireGuard.

2. UNDERLAY VS. OVERLAY NETWORKS

Understanding the distinction between underlay and overlay networks is essential for comprehending how GRE tunnels operate within a network architecture.

2.1 Underlay Network

The Underlay Network is the physical (or logical Layer 3) infrastructure that provides basic IP reachability between all network endpoints. It is the foundation upon which overlay technologies are constructed.

Core Components:

- Physical transmission media: fiber optic, copper, wireless links.
- Routing infrastructure: routers running OSPF, EIGRP, BGP, or IS-IS.
- Layer 2 switching fabric: Ethernet switches, VLANs.
- Public IP address space for inter-site reachability.

In the GRE lab topology, the underlay consists of:

- ISP Router providing transit routing between CAIRO and TANTA.
- Public subnets 20.1.1.0/24 (CAIRO-ISP link) and 30.1.1.0/24 (TANTA-ISP link).
- OSPF Area 0 distributing reachability between all three routers.

2.2 Overlay Network

The Overlay Network is a virtual network layer built on top of the underlay. It encapsulates original data packets inside new transport packets, allowing them to traverse the underlay while maintaining logical separation and the appearance of direct connectivity.

Common Overlay Technologies:

- GRE Tunnels (Generic Routing Encapsulation) — Protocol 47.
- IPsec Tunnels — provides encrypted overlay.
- DMVPN (Dynamic Multipoint VPN) — scalable hub-and-spoke with mGRE.
- VXLAN (Virtual Extensible LAN) — Layer 2 over Layer 3.
- MPLS VPN (L2VPN, L3VPN) — provider-managed overlays.

In the GRE lab, the overlay is the GRE Tunnel interface (Tunnel 0) configured on the CAIRO and TANTA routers, creating a virtual point-to-point link in the 100.1.1.0/24 subnet over the public Internet.

3. GENERIC ROUTING ENCAPSULATION (GRE)

GRE Protocol Summary

GRE (Generic Routing Encapsulation) is a tunneling protocol developed by Cisco and standardized in RFC 2784. It operates at Layer 3, carries IP Protocol Number 47, and is capable of encapsulating a wide variety of network layer protocols inside IP tunnels — enabling transport of private or non-routable traffic across public IP networks.

3.1 Technical Specifications

Parameter	Value / Description
IP Protocol Number	47
RFC Reference	RFC 2784 (Updated by RFC 2890)
OSI Layer	Layer 3 (Network Layer)
Supported Passenger Protocols	IPv4, IPv6, IPX, AppleTalk, IS-IS, and others
Delivery Protocol	IP (IPv4 / IPv6)
Tunnel Type	Point-to-Point (Logical)
Encryption Support	None (Plain GRE) — requires IPsec for encryption
Multicast / Broadcast Support	Yes (unique advantage over IPsec-only tunnels)
Routing Protocol Support	Yes — OSPF, EIGRP, RIP, BGP can traverse GRE tunnels
MTU Overhead	24 bytes (20-byte IP header + 4-byte GRE header, minimum)

3.2 GRE Packet Structure

When a packet is sent through a GRE tunnel, the original IP packet (the 'passenger' packet) is encapsulated with a GRE header and a new outer IP header (the 'transport' header). This is the core mechanism of GRE encapsulation:

Original IP Header	Original Payload (Data)	GRE Header	New IP Header (Outer)	Encapsulated GRE Packet
Src: 10.1.1.1 Dst: 192.168.1.1	TCP/UDP/Application Data	Protocol=0x0800 Version, Flags	Src: 20.1.1.2 Dst: 30.1.1.2	Travels over public Internet

GRE Header Field Details (minimum 4 bytes):

Field	Size	Description
Flags & Version	2 bytes	Control flags (C, R, K, S bits) and GRE version number (0 for RFC 2784)
Protocol Type	2 bytes	EtherType of the encapsulated (passenger) protocol. 0x0800 = IPv4, 0x86DD = IPv6
Checksum (optional)	2 bytes	Present when the C flag is set. Provides basic error detection
Reserved (optional)	2 bytes	Reserved field when checksum is present
Key (optional)	4 bytes	Present when K flag is set. Used to identify GRE flows (RFC 2890)
Sequence No. (optional)	4 bytes	Present when S flag is set. Enables packet ordering for delay-sensitive flows

3.3 GRE Encapsulation Process — Packet Flow

The following walkthrough describes the step-by-step packet flow when host 'cairo' (10.1.1.1) sends data to host 'tanta' (192.168.1.1) via the GRE tunnel:

Step 1 — Original Packet Originates at Source Host

The CAIRO VPCS generates an IP packet with Source IP: 10.1.1.1 and Destination IP: 192.168.1.1. The packet is forwarded to its default gateway (CAIRO Router, f1/0 = 10.1.1.10).

Step 2 — Routing Decision at CAIRO Router

The CAIRO router performs a routing lookup. Traffic destined to 192.168.1.0/24 is matched to a route via the Tunnel 0 interface (overlay route), not the physical f0/0 WAN interface.

Step 3 — GRE Encapsulation

The original packet (Inner IP Header + Data) is wrapped with a GRE Header (Protocol Type: 0x0800 for IPv4). A new outer IP Header is added with Src: 20.1.1.2 (CAIRO WAN) and Dst: 30.1.1.2 (TANTA WAN). The resulting encapsulated packet is forwarded to the ISP via f0/0.

Step 4 — Transit Through Underlay (ISP)

The ISP router sees only the outer IP header (20.1.1.2 → 30.1.1.2). It is unaware of the GRE encapsulation or the inner private payload. OSPF routing distributes reachability across all underlay routers.

Step 5 — GRE Decapsulation at TANTA Router

The TANTA router receives the encapsulated packet on its f0/0 interface. It identifies IP Protocol 47 (GRE), strips the outer IP header and GRE header, and recovers the original inner packet (Dst: 192.168.1.1). The packet is then forwarded out the f1/0 LAN interface to the TANTA VPCS.

Step 6 — Delivery at Destination

The TANTA VPCS receives the original, unencapsulated IP packet as if it were on the same local network as the Cairo host — achieving transparent LAN-to-LAN connectivity across the public Internet.

3.4 Advantages of GRE

- **Multi-Protocol Support:** Can tunnel non-IP protocols (IPX, AppleTalk) and IPv6 over IPv4 infrastructure.
- **Multicast & Broadcast Support:** Enables dynamic routing protocols (OSPF, EIGRP, RIP) to operate across the tunnel — a major limitation of plain IPsec.
- **Lightweight Header:** Minimum 4-byte GRE overhead is significantly smaller than other tunneling mechanisms.
- **Simple Configuration:** Requires minimal commands on Cisco IOS to establish a functional tunnel.
- **Protocol Agnostic:** Works with any passenger protocol that has an assigned EtherType value.
- **Scalable:** Can be combined with DMVPN for large-scale dynamic multi-point deployments.

3.5 Limitations of GRE

- **No Native Security:** GRE provides no encryption, authentication, or data integrity protection by itself.
- **Packet Overhead:** Minimum 24-byte overhead per packet reduces effective MTU (typically from 1500 to 1476 bytes).
- **MTU/Fragmentation Issues:** May cause TCP MSS clamping issues; requires `ip tcp adjust-mss` on tunnel interfaces.

- No QoS Awareness: Traffic classification and QoS markings on inner headers may not be preserved by default.
- Point-to-Point Only: Standard GRE is strictly point-to-point; multipoint requires mGRE (DMVPN).
- Security Dependency: Must be paired with IPsec (GRE over IPsec) for any production security requirement.

4. GNS3 LAB — NETWORK TOPOLOGY

4.1 Topology Diagram

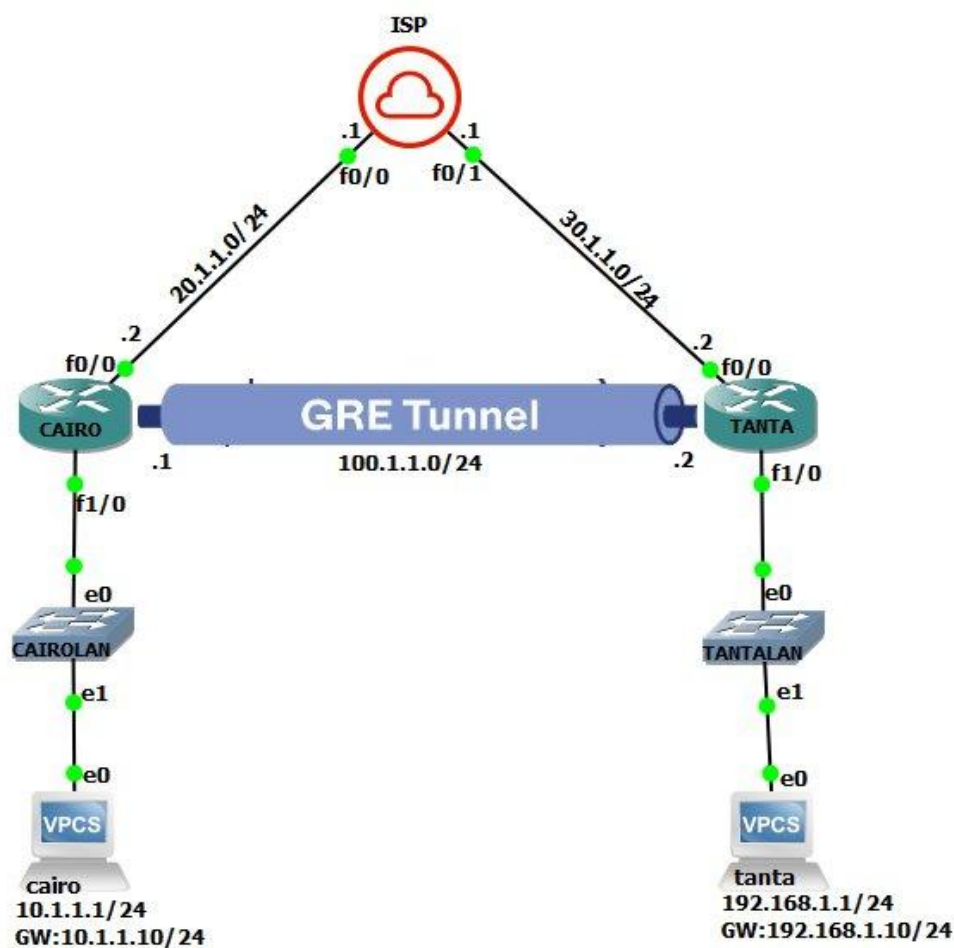


Figure 1: Site-to-Site GRE VPN Topology — CAIRO to TANTA via ISP

4.2 Topology Overview

This lab topology demonstrates a Site-to-Site GRE Tunnel connecting two geographically separated company branches — CAIRO and TANTA — across a simulated public Internet Service Provider (ISP) network. The topology illustrates the fundamental separation between the underlay (physical routing infrastructure) and the overlay (virtual GRE tunnel).

4.3 Underlay Network (Physical Path)

The underlay provides basic IP reachability between the CAIRO and TANTA tunnel endpoints across the ISP infrastructure.

- ISP Router: Simulates a public Internet service provider with two WAN-facing interfaces.
- CAIRO-ISP Link: Subnet 20.1.1.0/24 — CAIRO Router (20.1.1.2) connects to ISP (20.1.1.1).
- TANTA-ISP Link: Subnet 30.1.1.0/24 — TANTA Router (30.1.1.2) connects to ISP (30.1.1.1).
- Routing Protocol: OSPF Area 0 propagates routes between all three routers in the underlay.

4.4 Overlay Network (GRE Virtual Path)

The overlay provides virtual point-to-point connectivity between CAIRO and TANTA, making both LANs appear logically adjacent despite physical distance.

- GRE Tunnel Interface: Tunnel 0 on both routers — assigned subnet 100.1.1.0/24.
- CAIRO Tunnel Endpoint: 100.1.1.1 — sourced from f0/0 (20.1.1.2), destined to 30.1.1.2.
- TANTA Tunnel Endpoint: 100.1.1.2 — sourced from f0/0 (30.1.1.2), destined to 20.1.1.2.
- LAN Networks: CAIRO LAN (10.1.1.0/24) and TANTA LAN (192.168.1.0/24) reach each other via the tunnel.

5. IP ADDRESSING TABLE

The following table provides a comprehensive summary of all IP addresses, interfaces, and subnet masks used in this GRE VPN lab topology:

Device	Interface	IP Address	Subnet Mask / CIDR
CAIRO Router	FastEthernet 0/0 (WAN)	20.1.1.2	255.255.255.0 /24
CAIRO Router	FastEthernet 1/0 (LAN)	10.1.1.10	255.255.255.0 /24
CAIRO Router	Tunnel 0 (GRE)	100.1.1.1	255.255.255.0 /24
TANTA Router	FastEthernet 0/0 (WAN)	30.1.1.2	255.255.255.0 /24
TANTA Router	FastEthernet 1/0 (LAN)	192.168.1.10	255.255.255.0 /24
TANTA Router	Tunnel 0 (GRE)	100.1.1.2	255.255.255.0 /24
ISP Router	FastEthernet 0/0	20.1.1.1	255.255.255.0 /24
ISP Router	FastEthernet 0/1	30.1.1.1	255.255.255.0 /24
CAIRO VPCS (End Host)	NIC e0	10.1.1.1	255.255.255.0 /24
TANTA VPCS (End Host)	NIC e0	192.168.1.1	255.255.255.0 /24

6. DEVICE CONFIGURATIONS

6.1 IP Address Configuration

6.1.1 CAIRO Router

```

cairo(config)# interface FastEthernet0/0
cairo(config-if)# ip address 20.1.1.2 255.255.255.0
cairo(config-if)# no shutdown
cairo(config-if)# exit

cairo(config)# interface FastEthernet1/0
cairo(config-if)# ip address 10.1.1.10 255.255.255.0
cairo(config-if)# no shutdown
cairo(config-if)# exit

```

6.1.2 TANTA Router

```

tanta(config)# interface FastEthernet0/0
tanta(config-if)# ip address 30.1.1.2 255.255.255.0
tanta(config-if)# no shutdown
tanta(config-if)# exit

```

```
tanta(config)# interface FastEthernet1/0
tanta(config-if)# ip address 192.168.1.10 255.255.255.0
tanta(config-if)# no shutdown
tanta(config-if)# exit
```

6.2 Underlay Routing — OSPF Configuration

OSPF Area 0 is configured on all three routers to provide dynamic routing across the underlay. Only the WAN-facing public subnets are advertised into OSPF — the private LAN subnets are intentionally excluded from underlay routing.

6.2.1 CAIRO Router — OSPF

```
cairo(config)# router ospf 10
cairo(config-router)# network 20.1.1.0 0.0.0.255 area 0
cairo(config-router)# exit
```

6.2.2 ISP Router — OSPF

```
ISP(config)# router ospf 10
ISP(config-router)# network 0.0.0.0 0.0.0.0 area 0
ISP(config-router)# exit
```

6.2.3 TANTA Router — OSPF

```
tanta(config)# router ospf 10
tanta(config-router)# network 30.1.1.0 0.0.0.255 area 0
tanta(config-router)# exit
```

6.3 Overlay Configuration — GRE Tunnel

The GRE tunnel (Tunnel 0) is configured on both the CAIRO and TANTA routers to create the virtual overlay path. The tunnel source and destination reference the physical WAN interfaces to route through the ISP underlay.

6.3.1 CAIRO Router — GRE Tunnel

```
cairo(config)# interface Tunnel0
cairo(config-if)# ip address 100.1.1.1 255.255.255.0
```

```
cairo(config-if)# tunnel source FastEthernet0/0
cairo(config-if)# tunnel destination 30.1.1.2
cairo(config-if)# tunnel mode gre ip
cairo(config-if)# no shutdown
cairo(config-if)# exit
```

6.3.2 TANTA Router — GRE Tunnel

```
tanta(config)# interface Tunnel0
tanta(config-if)# ip address 100.1.1.2 255.255.255.0
tanta(config-if)# tunnel source FastEthernet0/0
tanta(config-if)# tunnel destination 20.1.1.2
tanta(config-if)# tunnel mode gre ip
tanta(config-if)# no shutdown
tanta(config-if)# exit
```

6.4 End-Host (VPCS) Configuration

6.4.1 CAIRO VPCS

```
cairo-vpcs> ip 10.1.1.1 255.255.255.0 10.1.1.10
Configuring IP: 10.1.1.1 | Mask: /24 | Gateway: 10.1.1.10
```

6.4.2 TANTA VPCS

```
tanta-vpcs> ip 192.168.1.1 255.255.255.0 192.168.1.10
Configuring IP: 192.168.1.1 | Mask: /24 | Gateway: 192.168.1.10
```

7. VERIFICATION & TESTING

After applying all configurations, the following verification commands and tests confirm that the GRE tunnel is operational and that end-to-end connectivity is established between the two LANs.

7.1 Tunnel Interface Status Verification

```
cairo# show interface Tunnel0
Tunnel0 is up, line protocol is up
```

```
Hardware is Tunnel
Internet address is 100.1.1.1/24
MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation TUNNEL, loopback not set
Keepalive not set
Tunnel source 20.1.1.2 (FastEthernet0/0), destination 30.1.1.2
Tunnel protocol/transport GRE/IP

tanta# show interface Tunnel0
Tunnel0 is up, line protocol is up
Internet address is 100.1.1.2/24
Tunnel source 30.1.1.2 (FastEthernet0/0), destination 20.1.1.2
Tunnel protocol/transport GRE/IP
```

7.2 End-to-End Connectivity Test (Ping)

```
cairo-vpcs> ping 192.168.1.1 -t
192.168.1.1 icmp_seq=1 ttl=62 time=12.342 ms
192.168.1.1 icmp_seq=2 ttl=62 time=10.891 ms
192.168.1.1 icmp_seq=3 ttl=62 time=11.203 ms
192.168.1.1 icmp_seq=4 ttl=62 time=10.987 ms
- Tunnel connectivity confirmed: CAIRO LAN <-> TANTA LAN
```

7.3 Wireshark Packet Analysis

Using Wireshark on the CAIRO-ISP link (20.1.1.0/24 subnet), you can observe GRE encapsulation in action:

- Outer IP Header: Source = 20.1.1.2, Destination = 30.1.1.2, Protocol = 47 (GRE).
- GRE Header: Protocol Type = 0x0800 (IPv4 passenger protocol).
- Inner IP Header: Source = 10.1.1.1 (cairo-vpcs), Destination = 192.168.1.1 (tanta-vpcs).
- Payload: ICMP Echo Request/Reply (or TCP/UDP application data).

Security Note

Because plain GRE provides NO encryption, the inner IP headers and payload are visible as clear text in Wireshark. This demonstrates why GRE is classified as a Trusted VPN rather than a Secured VPN. For production deployments, GRE should always be protected by IPsec (GRE

over IPsec) to encrypt the encapsulated payload.

8. SUMMARY & CONCLUSIONS

This document has presented a comprehensive examination of Generic Routing Encapsulation (GRE) as a site-to-site VPN tunneling technology, covering theoretical foundations, protocol mechanics, packet structure, and practical Cisco IOS configuration.

Key takeaways from this technical study:

- GRE is a versatile, lightweight, Layer 3 tunneling protocol (IP Protocol 47) capable of transporting virtually any passenger protocol across an IP network infrastructure.
- The underlay/overlay architecture is fundamental to understanding how GRE tunnels create virtual connectivity on top of existing physical routing infrastructure.
- GRE's support for multicast and broadcast traffic enables dynamic routing protocols (OSPF, EIGRP, BGP) to operate across the tunnel — a capability absent from plain IPsec tunnels.
- GRE provides no inherent security. For any production or security-sensitive deployment, GRE must be combined with IPsec to form a GRE over IPsec solution that provides encryption, authentication, and integrity protection.
- The 24-byte packet overhead introduced by GRE necessitates careful MTU planning and TCP MSS adjustment on tunnel interfaces to avoid fragmentation issues.

9. REFERENCES

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